

START

DECLARATIONS

 str employeeFirstName
 str employeeLastName
 str employeeId
 str validStatus
 str securityNotes[]
 num employeeNumberOfDependents
 num hoursWorked
 num grossPay
 num hourlyRate
 num calculatedStateTaxes
 num calculatedFedTaxes
 num overtimePay
 num netPay
 num overtimeHours
 inputfile employeeDatabase.dat

WHILE employeeFirstName NOT "zzz"

 employeeEntry()

 IF employeeFirstName = "zzz"

 BREAK

 ENDIF

 payrollCalculations()

 outputResults()

ENDWHILE

STOP

employeeEntry()

 BEGIN LOOP

 DISPLAY "Enter the employees first name or zzz to quit "

 INPUT employeeFirstName

 IF employeeFirstName NOT str

 "Error, please reenter employee first name"

 LOOP

 ELSE

 END LOOP

 ENDIF

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BEGIN LOOP
    DISPLAY "Enter the employees last name "
    INPUT employeeLastName
    IF employeeLastName NOT str
        DISPLAY "Error, please reenter employee last name"
        LOOP
    ELSE
        END LOOP
    ENDIF

```

```

BEGIN LOOP
    DISPLAY "Enter the employee's ID "
    INPUT employeeId
    IF employeeId NOT num OR length(employeeId) <> 5
        DISPLAY "Error, please reenter employee ID"
        LOOP
    ELSE
        OPEN employeeDatabase.dat
        hourlyRate = employeeDatabase[employeeId]
        IF hourlyRate not found
            DISPLAY error
            LOOP
        ELSE
            END LOOP
    ENDIF

```

```

BEGIN LOOP
    DISPLAY "Enter the employee's number of dependents "
    INPUT employeeNumberOfDependents
    IF employeeNumberOfDependents NOT num
        DISPLAY "Error invalid number, Please reenter the employee's number of dependents"
        LOOP
    ELSEIF employeeNumberOfDependents < 0
        DISPLAY "Error, invalid dependents"
        validStatus = "No"
        APPEND "Negative dependents" TO securityNotes[]
        LOOP
    ELSE
        END LOOP
    ENDIF

```

```

BEGIN LOOP
    DISPLAY "Please enter the number of hours worked "
    INPUT hoursWorked
    IF hoursWorked NOT num
        DISPLAY "Error, input not a valid number "
    ELSEIF hoursWorked > 60
        DISPLAY "Error, hours too high input wasn't within overtime hours"
        validStatus = "No"
        APPEND "Total hours exceed limit" TO securityNotes[]
    LOOP
    ELSE
        END LOOP
    ENDIF

```

RETURN

payrollCalculations()

```

IF hoursWorked <= 40 THEN
    grossPay = hoursWorked * hourlyRate
    overtimeHours = 0
    overtimePay = 0
ELSE
    overtimeHours = hoursWorked - 40
    overtimePay = overtimeHours * hourlyRate * 1.5
    grossPay = (40 * hourlyRate) + overtimePay
ENDIF
preTaxPay = grossPay
calculatedStateTaxes = grossPay * 0.056
calculatedFedTaxes = grossPay * 0.079
netPay = grossPay - calculatedStateTaxes - calculatedFedTaxes

```

RETURN

outputResults()

```

output "Employee Payroll Results"
output "Employee name: " + employeeFirstName + " " + employeeLastName
output "Employee ID: " + employeeId
output "Dependents: " + employeeNumberOfDependents
output "Hourly Rate: " + "$" + hourlyRate

```

```
output "Hours worked: " + hoursWorked
output "Overtime Hours: " + overtimeHours
output "Overtime Pay: " + "$" + overtimePay
output "Gross Pay: " + "$" + grossPay
output "Federal Tax: " + "$" + calculatedFederalTaxes
output "State Tax: " + "$" + calculatedStateTaxes
output "Net Pay: " + "$" + netPay
output "Valid: " + validStatus
output "Security Note: " + securityNotes[]
```

RECORD results TO spreadsheet

RETURN